

ME4513 - Industrial Products and Machinery Development

**Development of A Walking Mechanism for
Abelow-Knee Prosthetic Leg to Fatigue load testing**

By

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1. INTRODUCTION

Prosthetic leg developers and manufacturers require reliable methods to evaluate the performance, durability, and safety of prosthetic legs prior to human use. Many existing testing approaches are either manual or dependent on human subjects, which introduces safety risks, ethical concerns, and inconsistencies in test results. Furthermore, manual testing methods lack repeatability and make accurate quantification of fatigue life difficult. These limitations highlight the need for a standardized mechanical testing solution.

Accordingly, there is a clear consumer demand for a mechanical testing system capable of replicating human walking motion and applying realistic body loads in a controlled, repeatable, and safe manner. Such a system enables early identification of potential fatigue failures, reduces development time and cost, and improves the overall reliability of prosthetic legs before clinical trials.

This project aims to design and develop a mechanically driven testing mechanism for evaluating a prosthetic leg. The primary objective is to create a system that produces a controlled and repeatable walking motion while ensuring stable, natural, and safe operation. Emphasis is placed on simplicity, cost-effectiveness, and mechanical reliability. The mechanism is designed to reproduce the major phases of the human gait cycle, such as heel-strike, stance, toe-off, and swing, while accommodating a range of applied load conditions. The developed mechanism is intended to evaluate the mechanical performance of the prosthetic leg under realistic walking conditions, including structural durability, torque tolerance, angular and vertical loading, gait simulation accuracy, and fatigue life.

2. CONSUMER NEEDS

1. Safe pre-clinical testing

A testing system that allows prosthetic legs to be evaluated without exposing human subjects to risk.

2. Repeatable and consistent testing conditions

The ability to perform tests under identical motion and loading conditions to ensure reliable and comparable results.

3. Realistic gait simulation

A mechanism capable of reproducing the major phases of the human gait cycle, including heel-strike, stance, toe-off, and swing.

4. Accurate load application and measurement

The ability to apply and measure body weight equivalent loads and joint forces relevant to real walking conditions.

5. Fatigue life evaluation

A system that enables long duration cyclic testing to assess durability and fatigue failure of prosthetic legs.

6. Adjustability and versatility

Capability to accommodate different prosthetic sizes, configurations, and load levels.

7. Cost-effective and simple design

A mechanically simple system that can be manufactured and operated at low cost compared to advanced robotic test rigs.

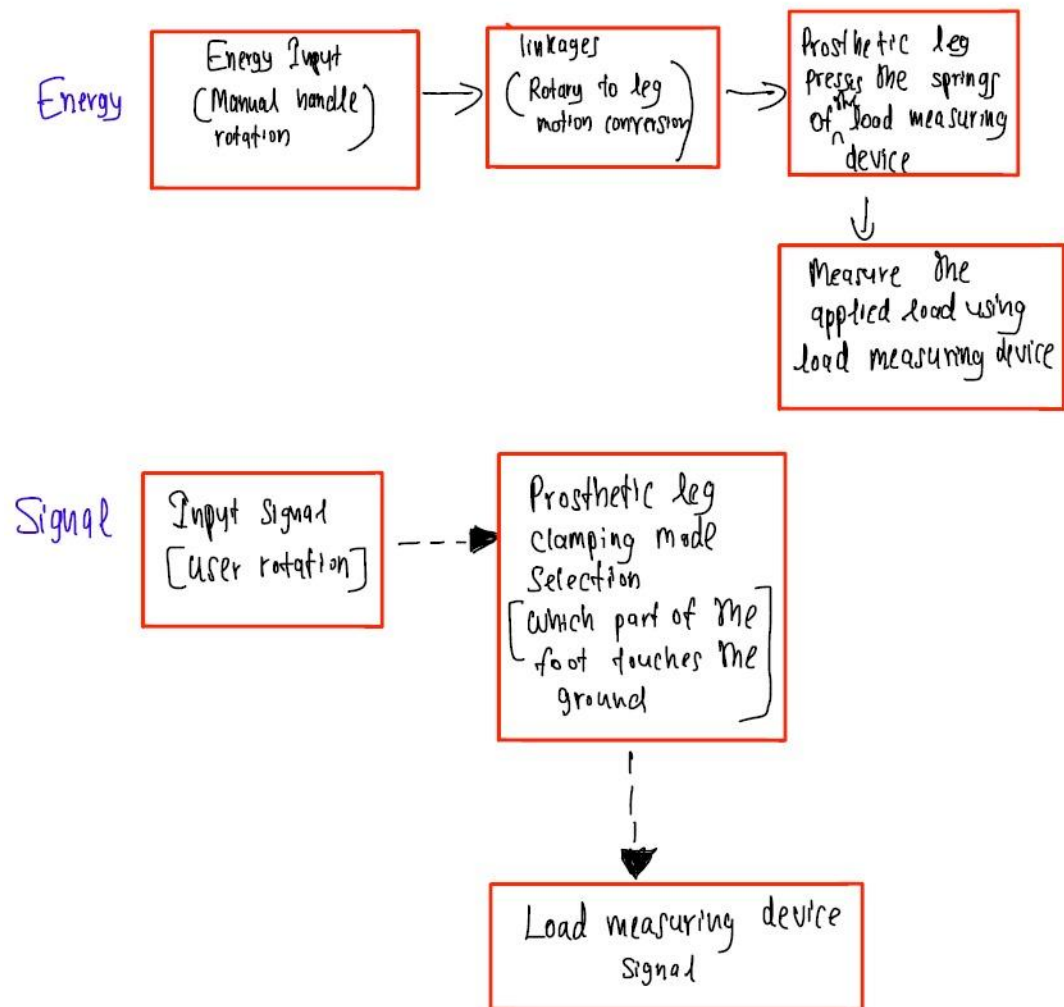
8. Ease of setup and operation

A user friendly system that allows quick mounting, testing, and replacement of prosthetic legs.

9. Durability

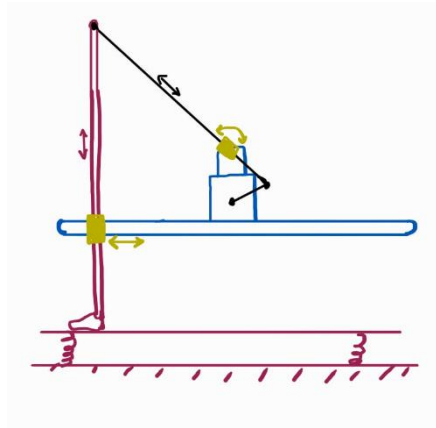
The mechanism should be strong enough to perform continuous operation for extended periods without structural failure or significant wear.

3. FUNCTIONAL BLOCK DIAGRAM



4. DESIGN CONCEPTS

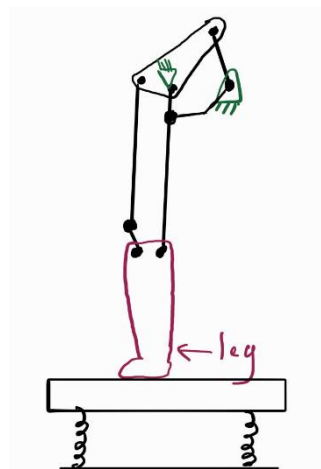
1. Concept 1



This mechanism uses multiple linkages to achieve the motion of the leg. Circular part gets the rotational movement from externally. The leg has more horizontal movement, so that the platform also designed eventually.

The prosthetic leg contact surface is spring loaded to allow adjustable force application. By varying the spring height different contact forces can be achieved in accordance with Hooke's law ($F = kx$). This arrangement enables more natural foot ground contact behavior during testing.

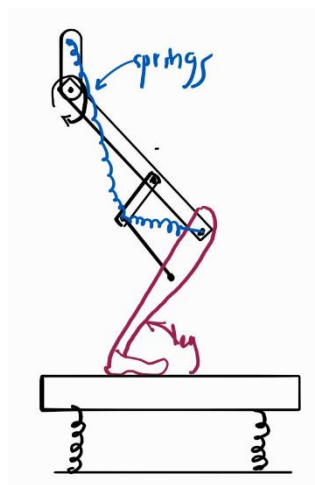
2. Concept 2



The leg motion is generated through a combination of interconnected linkages. An external source provides rotational input to a circular driving component, while a triangular element is included to improve structural strength and stability.

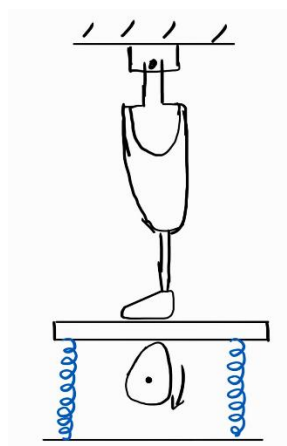
The interface where the prosthetic leg makes contact is supported by a spring loaded system, allowing the applied force to be adjusted. By changing the spring compression different contact forces can be produced in accordance with Hooke's law ($F = kx$). This configuration helps achieve a more realistic simulation of foot ground interaction during testing.

3. Concept 3



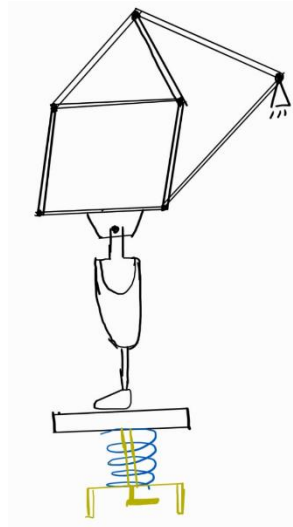
Here is the leg has multiple linkages. Addition to these linkages there are several springs to control the movement of the mechanism. The crank located at the top is rotating with the externally applied force. The stage of this mechanism is also same as the previous concept's stage.

4. Concept 4



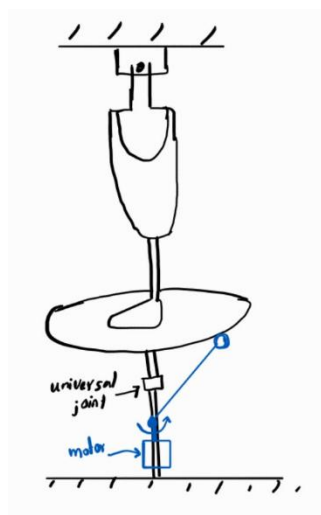
c Real mechanism happens of the stage. Here the stage is holding by four springs while crank is rotating to give the forces similar to walking.

5. Concept 5



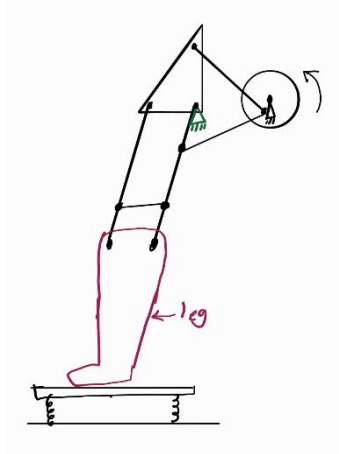
Here the mechanism has multiple number of linkages. Leg is mounted to the bottom most link. External rotating force is applied to the pivoted point. Stage of this concept if mounted to the spring. The height of the spring can be vary with the axle going through the spring. This can be done by the hand.

6. Concept 6



Here the leg mounter is pivoted at the top. Stage is mounted to the floor with a universal joint. Then a rotating angled rod which is mounted to the motor is contacting with the stage. This gives forces in all angles to the foot.

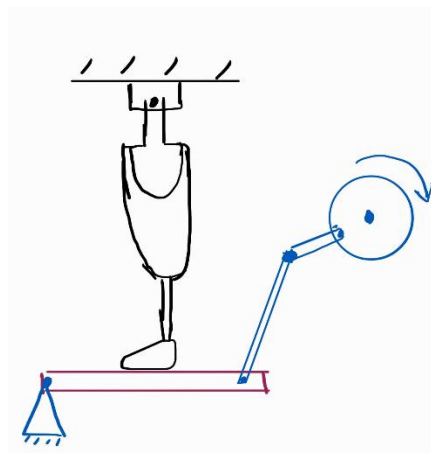
7. Concept 7



This mechanism uses multiple linkages to achieve the motion of the leg. Circular part gets the rotational movement from externally. In the mechanism there is a triangular part for additional stability of the strength of the structure.

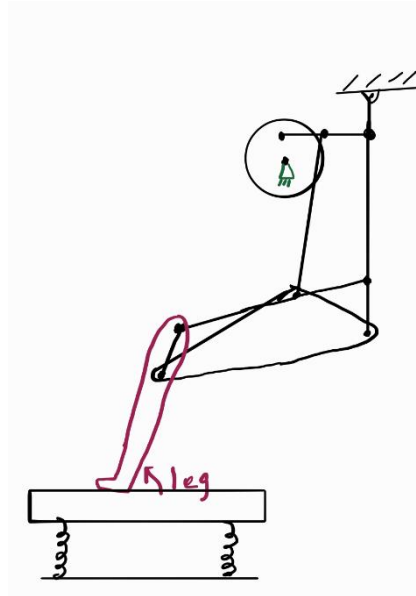
The prosthetic leg contact surface is spring loaded to allow adjustable force application. By varying the spring height different contact forces can be achieved in accordance with Hooke's law ($F = kx$). This arrangement enables more natural foot ground contact behavior during testing.

8. Concept 8



Here the leg mounter is pivoted at the top. The stage is pivoted from one end. The other end is connected to a linkage. This linkage gets its required motion by the crank, as the image. When the crank is rotating state exerts forces to the leg periodically like when walking.

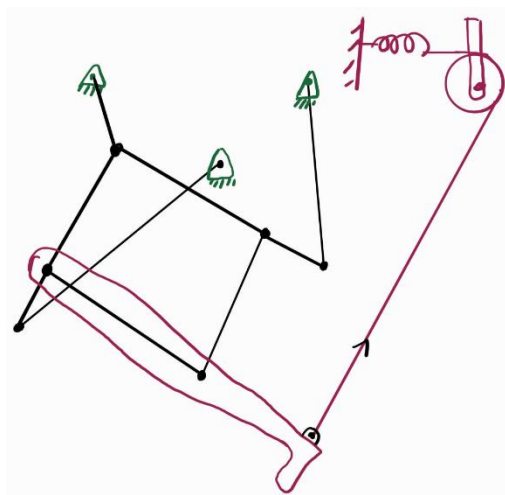
9. Concept 9



The mechanism employs a system of interconnected linkages to reproduce the required leg motion. Rotational input is supplied externally to a circular driving element, while a triangular component is incorporated to enhance structural rigidity and overall stability.

The surface that interfaces with the prosthetic leg is supported by a spring loaded arrangement, enabling controlled variation of the applied force. Adjusting the spring compression allows different force levels to be generated based on Hooke's law ($F = kx$), thereby providing a more realistic representation of foot ground contact during experimental evaluation.

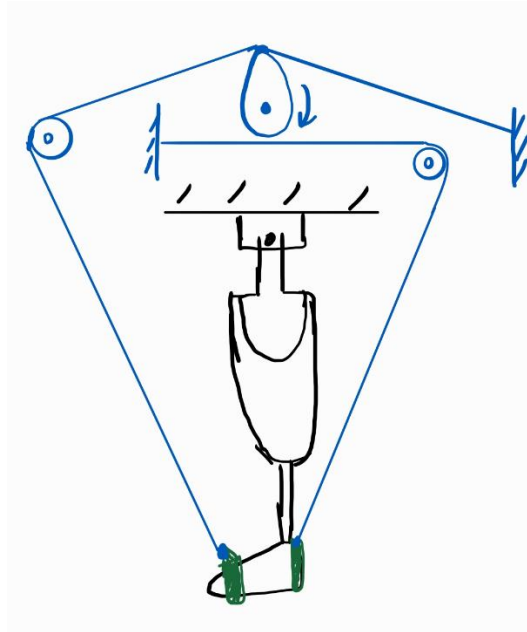
10. Concept 10



Bottom of the leg is connected to the string by a strap. The string is then goes through a coupling which is then connected to a spring. This gives resistance to each step. Here the shortest linkage is rotated by external force.

9

12. Concept 12



Here, the leg mount is pivoted at the top, allowing rotational movement similar to a natural leg motion. Two strings are attached to the leg using straps, and these strings are driven by a rotating crank positioned at the top. As the crank rotates, tensile forces are generated in the strings, which in turn drive the leg through a cyclic motion, replicating the dynamic forces experienced during walking.

5. CONCEPT SCORING

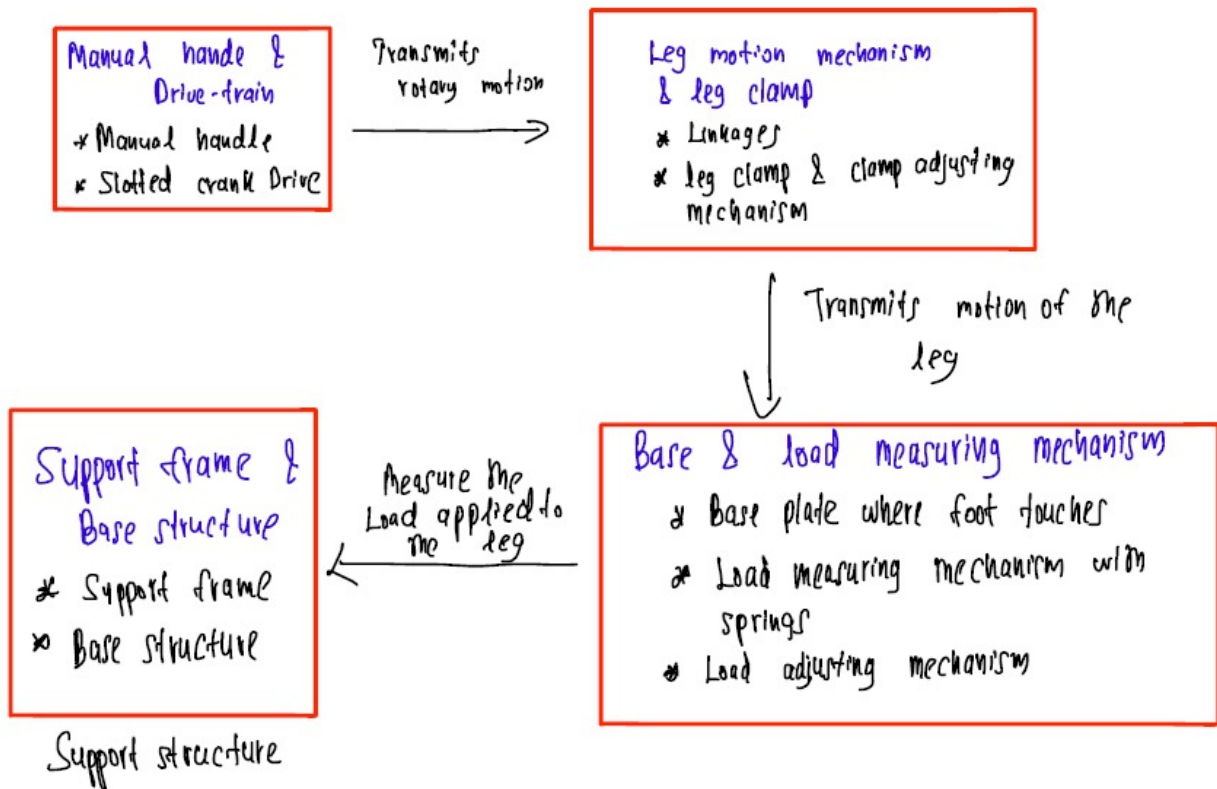
Selection Criteria	Weight	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12
Range of Motion	35%	3	2	4	2	4	4	4	3	2	4	2	2
Stability	20%	2	2	3	3	3	3	4	3	4	3	4	3
Force Transmission	20%	4	3	3	3	3	3	4	3	4	3	2	3
Compactness & Weight	10%	2	2	3	3	3	3	4	3	4	3	3	4
Simplicity / Links	5%	4	4	4	5	4	5	3	2	3	3	2	5
Manufacturability	4%	4	4	4	4	4	4	4	3	4	3	2	5
Durability	3%	3	3	3	4	3	3	3	2	3	3	3	2
Ease of Adjustment	3%	3	3	3	3	3	3	3	3	3	3	5	3
Weighted Total	100%	2.95	2.37	3.46	2.76	3.46	3.51	3.89	2.89	3.22	3.42	2.67	2.74
New Rank		6	12	3	8	3	2	1	7	5	4	11	9

The selection process evaluates twelve design concepts (D1–D12) using a Weighted Decision Matrix to identify the most effective solution. This method prioritizes critical functional requirements by assigning 75% of the total score to three primary drivers: Range of Motion (35%), Stability (20%), and Force Transmission (20%). Secondary factors like compactness, simplicity, and manufacturability account for the remaining 25% of the evaluation.

Based on these priorities, Design 7 emerged as the top-ranked concept with a score of 3.89, having achieved high marks across all three primary performance categories. Design 6 followed in second place with a score of 3.51, while Design 3 and Design 5 tied for third. Conversely, designs like D2 and D11 ranked lowest because they failed to perform well in the high-weight functional areas, despite their scores in simpler categories.

6. PHYSICAL CHUNKS OF FINAL DESIGN

Physical chunks



7. FINAL DESIGN SELECTED FOR PROTOTYPING

After completing the concept screening processes, the most suitable design was selected based on its balanced performance across motion accuracy, stability, manufacturability, and simplicity. The selected concept was then developed into a detailed 3D model using SolidWorks, where all functional components, linkages, joints, and interfaces were defined and verified. Following the detailed design stage, a physical prototype was fabricated incorporating all the intended functional features of the system, enabling practical evaluation and validation of the design.

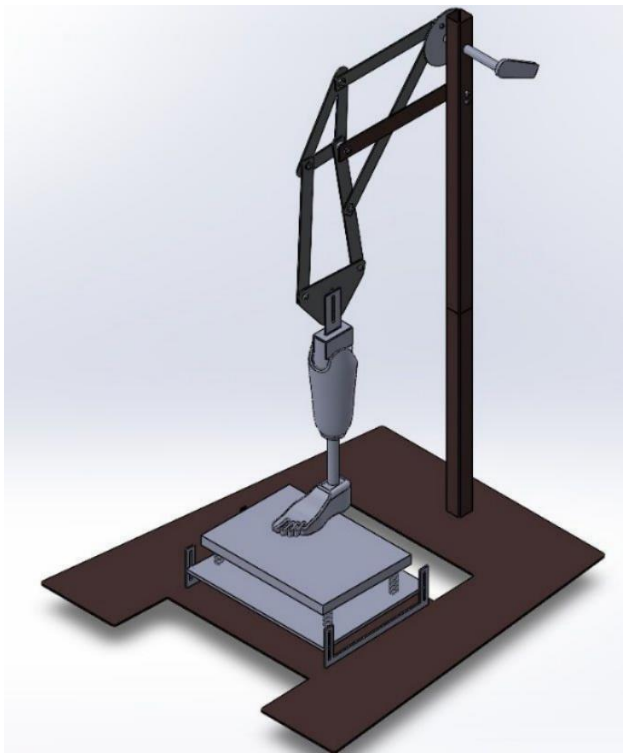


Figure 2: 3D model



Figure 1: Prototype

The final design is a mechanically driven prosthetic leg fatigue testing mechanism developed to simulate realistic human walking motion under controlled loading conditions. The system is mounted on a rigid vertical support frame that provides structural stability and maintains proper alignment during continuous cyclic operation. The overall design emphasizes simplicity, repeatability, safety, and cost effectiveness, making it suitable for pre-clinical evaluation of prosthetic legs.

Motion mechanism

The walking motion is generated using a planar linkage mechanism driven by a rotary input. The rotary motion, supplied through a crank at the top of the frame, is converted into a leg like trajectory by the linkage assembly. Carefully selected link lengths and pivot locations enable the mechanism to produce knee flexion during the swing phase and leg extension during the stance phase. This motion closely approximates real human leg motion at the joint level and allows reproduction of the major phases of the human gait cycle, including heel-strike, stance, toe-off, and swing. While the mechanism does not replicate muscular action, it accurately reproduces the kinematic behavior relevant to fatigue testing.

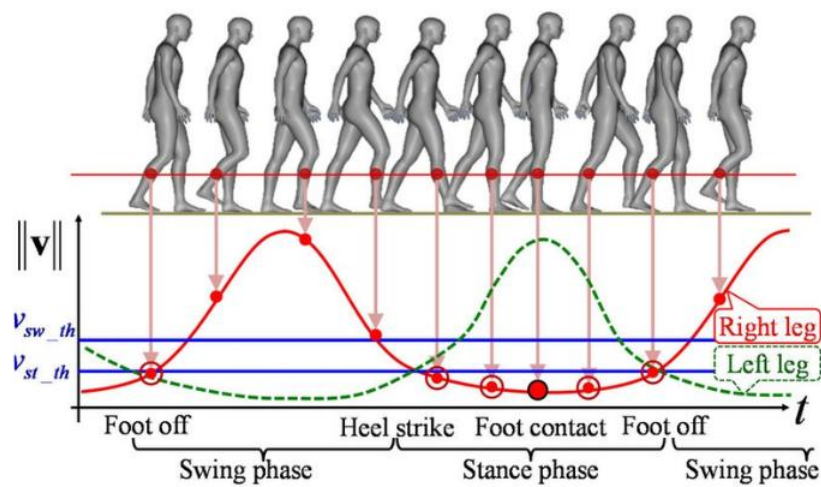


Figure 3: Real leg motion

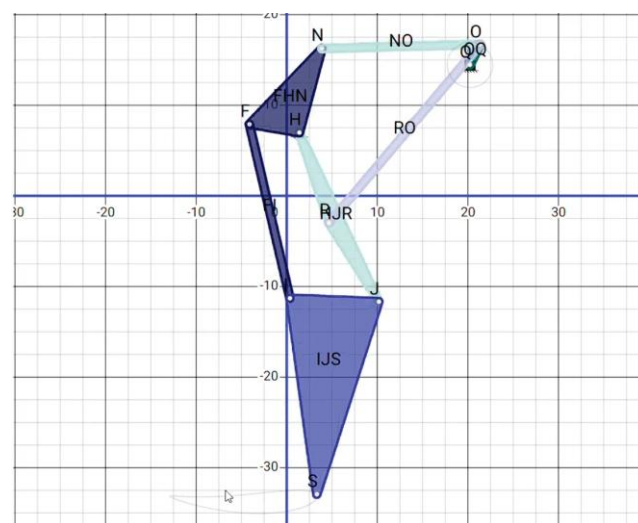


Figure 4: Selected linkage design showing link lengths chosen to closely approximate real human leg motion.

Mounting mechanism

The prosthetic leg is mounted to the mechanism using a attachment system similar to that used by human users. This mounting approach ensures realistic boundary conditions and load transfer between the mechanism and the prosthetic leg. To improve alignment and stability. This configuration allows secure fixation while enabling easy installation and removal of different prosthetic legs for testing.

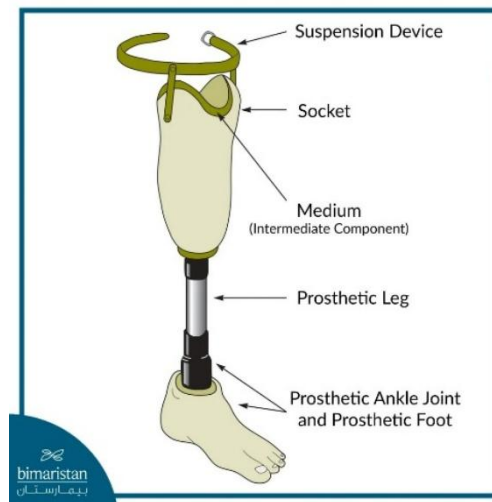


Figure 5: Prosthetic leg mounting mechanism which plan to be used in actual design.
This is not included in the prototype.

The leg mounting plate can have adjustable height by changing the holes used for the nuts and bolts, allowing it to accommodate various prosthetic legs.

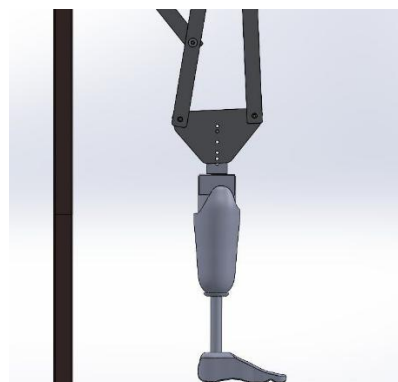


Figure 6: Adjustable height leg mounting plate

Loading mechanism

Body-weight-equivalent loading is applied vertically through the prosthetic leg to simulate realistic walking loads. When the prosthetic foot contacts the ground, the load is transferred to a base plate supported by four identical compression springs. As the applied load increases, the springs compress proportionally. The applied force is calculated using Hooke's Law by measuring the spring compression. This spring-based load measurement method provides a simple, reliable, and cost-effective way to quantify the load applied to the prosthetic leg during testing.

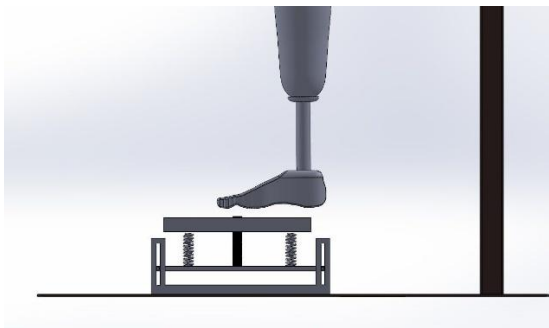


Figure 8: Loading mechanism in 3D model



Figure 7: Loading mechanism prototype

The applied load on the prosthetic leg can be adjusted by changing the height of the base plate. The design and the prototype of the base legs that allows for easy adjustment to change the base height for that. When the base height is increased, the prosthetic leg compresses the springs by a greater amount during contact, resulting in an increased applied load. Conversely, reducing the base height decreases spring compression and the applied load.

In the current prototype, spring compression is measured manually using a linear scale. Although this method is sufficient for preliminary testing, it introduces minor measurement uncertainties. As a future improvement, the integration of an LVDT (Linear Variable Differential Transformer) is planned to enable accurate, real time measurement of spring displacement and automated data acquisition.

The system is designed to operate in a controlled and repeatable manner, allowing long-duration cyclic testing for fatigue life evaluation. The separation of motion generation, load application, prosthetic mounting, and structural support into distinct functional modules improves reliability, simplifies fabrication, and allows future upgrades such as motorized actuation and variable load profiles.

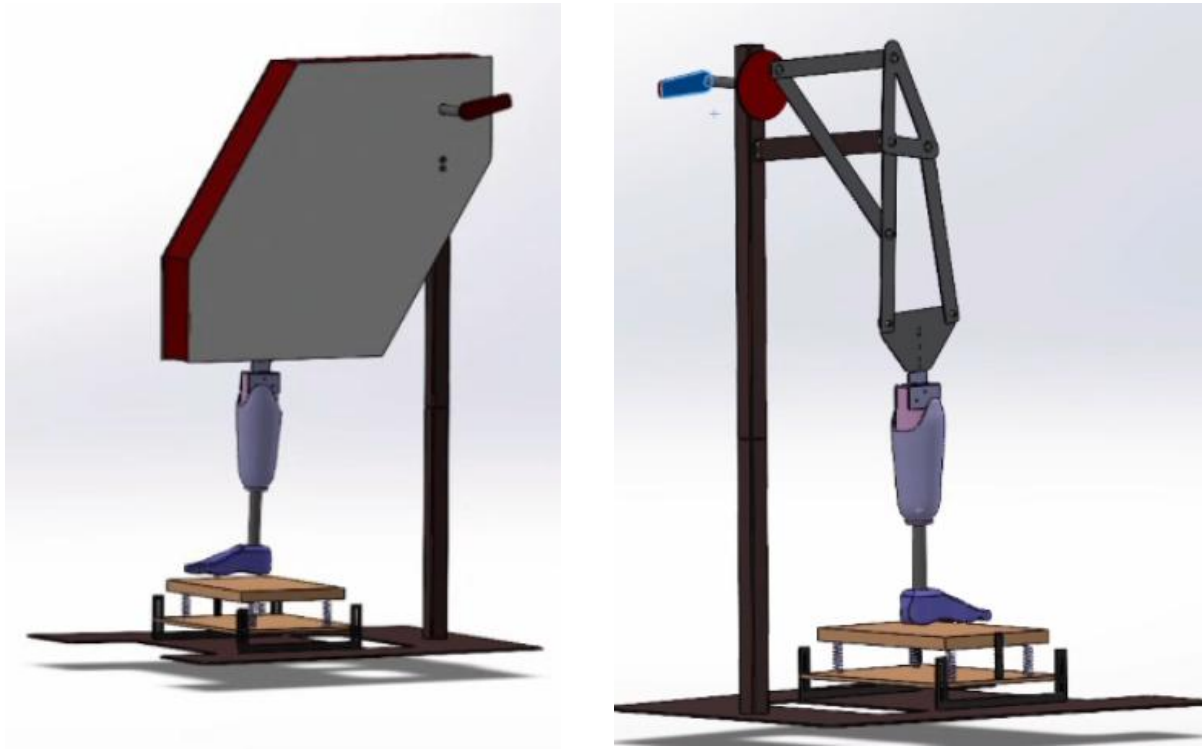


Figure 9: Final 3D model

Overall, the final design provides a safe, repeatable, and mechanically robust platform for evaluating the durability, load response, and fatigue behavior of prosthetic legs under realistic walking conditions, fulfilling the identified consumer needs and project objectives.

Demonstrating That the Final Design Fulfills Consumer Needs

Safe Pre-Clinical Testing

The prosthetic leg is mounted on a rigid mechanical frame and operated entirely without human involvement. All loads and motions are generated mechanically, eliminating risk to users while allowing extensive fatigue testing.

Repeatable and Consistent Testing Conditions

The mechanism is driven by a fixed kinematic linkage with defined pivot locations and link lengths. This ensures that each cycle produces the same motion path and joint angles. Load application is independent of motion, allowing consistent repetition over thousands of cycles.

Realistic Walking Motion Simulation

The linkage system converts rotary input into a leg-like trajectory that closely approximates human joint-level motion. The mechanism produces controlled knee flexion during swing and leg extension during stance, replicating key gait phases such as heel-strike, stance, toe-off, and swing.

Realistic Load Application

Vertical load is applied along the natural load path of the prosthetic leg, similar to how body weight is transferred from the hip to the foot during walking. This ensures realistic stress distribution within the prosthetic structure.

Accurate Load Measurement

The base is supported by four calibrated compression springs. When the prosthetic foot contacts the base, spring compression occurs. Using Hooke's Law, the applied force is calculated from measured spring deflection.

This is simple and cost-effective reliable measurement. We can improve accuracy upgrading path to LVDT for real-time data

Fatigue Life Evaluation

The mechanically driven linkage and robust frame allow continuous cyclic operation. The absence of complex electronics improves reliability during extended testing, making the system suitable for fatigue failure evaluation.

Compatibility with Real Prosthetic Attachment

The prosthetic leg is attached using a standard system, replicating how a human user wears the prosthesis. This ensures realistic load transfer and boundary conditions during testing.

Adjustability and Versatility

The modular structure allows easy adjustment of load levels, prosthetic mounting, and motion parameters. Different prosthetic legs can be installed without redesigning the system.

Cost-Effective and Simple Design

The design avoids complex actuators, sensors, or robotic systems. It uses standard mechanical components such as linkages, bearings, springs, and structural members, significantly reducing cost.

Manufacturability

The design was developed with manufacturability in mind, using processes that are efficient, precise, and accessible. Key components are fabricated using laser cutting, which ensures high dimensional accuracy and clean edges. Assembly is achieved through TIG welding for permanent joints and nut-and-bolt connections for parts that require disassembly or adjustment. This combination of fabrication and joining techniques simplifies the manufacturing process, reduces production time, and allows for easy maintenance or modifications.

Why This Was the Best Design

During the concept development phase, multiple design alternatives were explored, each with different strengths and weaknesses. After thorough evaluation, this design emerged as the optimal solution because it strikes a well-balanced compromise between several critical engineering factors: motion accuracy, structural stability, manufacturability, load realism, and cost-effectiveness. Other concepts fell short in one or more areas they were either too complex to manufacture reliably, had poor force transmission leading to inaccurate results, or introduced unnecessary points of potential failure. By carefully balancing these factors, this design ensures functional effectiveness while remaining practical for production and testing.

For a fatigue testing system, reliability and repeatability are more valuable than complexity. This design uses a simple kinematic chain with a minimal number of joints, which reduces friction, wear, and maintenance requirements. Fewer moving parts mean fewer sources of error and failure, directly improving the consistency of test results. A clear, strong principle underpins this approach: simplicity directly enhances reliability and data consistency, which is essential when testing prosthetic legs over repeated load cycles.

Another key advantage of this design is the clarity of its force transmission path. The applied motion travels through a well-defined sequence:

input motion → linkage → prosthetic leg → spring base → frame.

This straightforward path allows engineers to easily validate the system, understand stress distribution, and accurately assess fatigue life. A clear force flow not only simplifies analysis but also increases confidence in the test data by reducing uncertainties caused by unpredictable or indirect load paths.

In conclusion, this design successfully addresses all identified consumer needs by providing a safe, repeatable, realistic, and cost-effective platform for prosthetic leg fatigue testing. Its combination of balanced engineering trade-offs, simplicity for enhanced reliability, and transparent force flow makes it the most suitable concept for prototyping and further development, ensuring both functional effectiveness and practical manufacturability.

8 ISSUES IDENTIFIED AND PROPOSED DESIGN MODIFICATIONS

Identified Issues

1. Lack of relative motion between leg and base

Current design does not replicate the natural motion between the prosthetic leg and the ground. In real-life gait, the foot moves relative to the ground.

2. Manual load measurement is inefficient

Using a linear scale to measure load requires significant human effort. Measurement process is slow and prone to human error, reducing repeatability.

3. Fixed load profile limitations

Current system applies to a static or pre-defined load that does not fully represent dynamic forces experienced during walking. Important gait phases like variable heel impact and push-off are not accurately captured.

Proposed Improvements

1. Introducing relative motion between leg and ground

Implement a treadmill or treadwheel mechanism to replicate realistic foot-ground interaction. Allow natural gait phases (heel strike, mid-stance, toe-off) to occur under load.

2. Automated real-time load measurement

Integrate an LVDT (Linear Variable Differential Transformer) to measure spring compression continuously. Provides accurate, real-time data on force transmission without manual measurement.

3. Motorized drive system for automated cycling

Replace manual actuation with a programmable motor system. Enables continuous, consistent cycling for long-duration fatigue tests.

4. Variable load application

Implement adjustable load mechanisms using springs, cams, or pneumatic systems. Simulate realistic dynamic gait profiles rather than fixed loads.

5. Enhanced data acquisition and control

Integrate sensors for displacement, force, and motion tracking. Use software-controlled inputs for precise load profiles and automated test sequencing.